

On Predicting Success in Training for Men and Women: Examples From Marine Corps Clerical Specialties

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The presence of differences between prediction systems for men and women is investigated through a detailed study of clerical occupational specialties in the U.S. Marine Corps. When various measures of ability were used to predict success of recruits in training, sizeable differences in regression equations were found between men and women. The study showed that selected deletion of extraneous Armed Services Vocational Aptitude Battery (ASVAB) variables maintained overall predictive efficiency but did not entirely remove the differences between subgroup regressions. When the attainment of a high-school diploma was considered, however, subgroup differences between predicted scores were substantially reduced. Implications of these empirical results for the general problem of military personnel selection are discussed.

The literature in personnel selection and classification is replete with examples of both situationally specific and group-differentiated predictions of job performance. Unique characteristics of tasks peculiar to a particular job or job family have been thought to moderate the relation between a global set of predictors and on-the-job performance (cf. Ghiselli, 1966). Situational specificity in the prediction of success in training for military occupations, for example, has been a focus of attention for test validation at least since Stewart's (1947) analysis of occupational differences in scores from the Army General Classification Test. Accordingly, validation research performed in the military since that time has concentrated on the relation between test scores and training success within specific occupational groups (e.g., Valentine, 1977). Recent research in the area of validity generalization, however, has presented a strong challenge to specificity of prediction with respect to job setting.

This article has a slightly different, although similar, focus. It addresses the general issue of prediction differences for men and women by means of a detailed analysis of selected job classifications in the clerical specialty area of the U.S. Marine Corps. Our particular interest lies in the identification of differences between predicted training success for men and women on the basis of Armed Services Vocational Aptitude Battery (ASVAB) composite scores and subtests typically used for selection. The search for alternative predictive composites that yield equally accurate (in the sense of mean square error), nondifferentiated regression equations is another concern of this analysis.

The study of different prediction systems for men and women has a growing history in Air Force technical training programs.

Gordon (1953) was the first to find stable differences between the regression equations for men and women: Women's performance was consistently underpredicted in this early study. In a more recent validation study of composites used for predicting performance in Air Force mechanical training courses, Valentine (1977) found substantial differences between the predicted scores on the basis of regression equations for men and women in nine courses. For example, differences between criterion means observed for women and those predicted from regression equations on the basis of a sample of men participating in the same training programs ranged from .10 to .43 standard deviations for women (Linn, 1984). For the nine training courses examined, mean performance of women was underpredicted by an average of .28 standard deviations.

In contrast to the large discrepancies noted by Valentine (1977), Webster, Booth, Graham, and Alf (1978) found limited evidence supporting separate prediction equations for men and women in Navy hospital training schools. These authors examined the impact of a common prediction equation on the number of false-positive and false-negative decisions made for various cut-scores on the predictors. They found significant differences between men and women in the proportion of recruits misclassified, with men showing a higher proportion of misclassifications because of underprediction of their criterion scores using a common equation. Although the criterion used by Webster et al. was different, the contrast that their study poses with the previous Air Force results makes it reasonable to expect differences between regressions for men and women to be, at least to some extent, program specific. We return to this observation in discussion of the results presented below.

In a study of the regression of undergraduate grades on an American College Testing Program (ACT) battery, Gamache and Novick (1985) investigated the presence of male-female differentiated prediction for selected college programs (liberal arts, business, medicine, etc.). They identified differences between the predictions of grade-point average from ACT scores to be as much as .45 standard deviations for women on the crite-

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Table 1
Clerical Specialties and Corresponding Sample Sizes

Specialty	Total	Men		Women	
		N	%	N	%
Administrative	1,287	841	65	446	35
Personnel	170	125	74	45	26
Unit diary	148	96	65	52	35
Basic supply stock	955	848	89	107	11
Personal financial records	326	263	81	63	19
Aviation operations	233	190	82	43	18
Aviation maintenance administration	200	166	83	34	17
Aviation supply	458	408	89	50	11
Communications center	656	575	88	81	12
Totals	4,433	3,512	79	921	21

tion when the within-group regressions were evaluated at the mean predictor score for women. Again, the effect was to underpredict the criterion performance of women in most programs and with most variables. Similar tendencies for equations derived from samples of men to underpredict the performance of women have been observed in educational settings by Linn (1973), ACT (1973), and Breland and Griswold (1982). The findings of Breland and Griswold are noteworthy in that underprediction of performance by women was consistent throughout the predictor score scales and was documented through traditional regression techniques as well as a contingency table analysis.

Method

Data Source

Data for this report came from validation files used in a study of the Armed Services Vocational Aptitude Battery (ASVAB), Forms 6 and 7 (Sims & Hiatt, 1981). We limited consideration to recruits assigned to one of nine training courses classified as clerical specialties. Table 1 lists these clerical specialties and the sample sizes that were available for analysis. The proportion of women in the total sample ranged from .10 for Basic Supply Stock Clerks to .35 for Administrative and Unit Diary Clerks. It should be noted that the sample sizes in several of the training groups were smaller than would be desirable in most validation settings. We recognize this to be a problem both from the point of view of having large chance errors to cope with and from that of lacking power to reject false hypotheses, and we suggest that the ensuing analyses be interpreted with due regard to the concerns of each point of view.

Predictor variables of special interest in this study were the ASVAB Clerical Composite (CL) used for selection and the four ASVAB subtests from which the Clerical Composite is constructed. These subtests are Arithmetic Reasoning (AR), Word Knowledge (WK), Attention to Detail (AD), and an attitudinal measure called the Attentiveness Scale (CA). All ASVAB variables except CA were transformed to standard scores using the corrected conversion tables from Forms 6 and 7. This placed the composite on a scale with a mean of 100 and a standard deviation of 20 and the subtests on scales with means of 50 and standard deviations of 10. Because no standard scale was ever developed for CA, its scores range from 0 to 20. The available criterion variable was final course grade (FCG) in training. These scores are reported on a nominal

scale of 0–100, with values for most training courses falling between 70 and 100.

Procedure

The distributions of ASVAB predictors and course-grade criterion were first examined for ceiling and floor effects that might jeopardize the linearity and homoscedasticity requirements of the least squares regressions and for outliers. In several training courses, observations were deleted because of extreme scores on the criterion—Sims and Hiatt (1981) noted that extremely low criterion scores for these courses appeared to be arbitrarily determined. Note that there was no tendency for either men or women to be deleted from any of the samples with greater frequency. Area transformations of the criterion variable were also considered in situations where heteroscedasticity appeared to be a concern; however, no such transformations had a marked effect on the joint criterion-predictor distributions. Separate least squares regression equations for predicting training success were determined for the clerical composite, CL, and for each of the subtests of which it is composed. The subtests were also considered as multiple predictors of training success. Although the data available for analysis are clearly subject to the effects of range restriction on the independent variables, no attempts to remove these effects were considered in this study. One justification for this was the focus in this study on regressions rather than correlations; these are not affected by explicit selection.

Preliminary tests of null hypotheses of equal within-group slopes and intercepts were conducted using conventional techniques. For these tests an alpha level of .10 was established for rejection, the rationale being that Type II errors should be lessened in the study of differential prediction. In addition to these tests, regions of the predictor score scale where important differences between regression equations existed for men and women were identified using the Johnson-Neyman technique. This technique identifies values on the score scale where the 100 (1 – p) % prediction interval for differences between predicted scores does not include zero. A concise description of the Johnson-Neyman procedure is provided by Pottoff (1983).

A search for alternative predictive composites was guided by the approach described in Gamache and Novick (1985). This is basically an iterative search for a set of predictors (or a weighted composite of predictors) providing accuracy sufficient for administrative purposes while minimizing the proportion of individuals affected by any differential prediction that might exist. It is important to consider any such search as an exploratory analysis, whose results would ideally be cross-validated through additional sampling from these job categories.

Results

Within-group means and standard deviations for recruits in the nine courses are reported in Table 2. Although the distributions of composite scores varied from course to course, there seemed to be a marked tendency for women to have higher means on the clerical composite. Mean differences between men and women ranged from 7 to 15 scaled score points, with women scoring an average of 9 points higher than men across the nine schools. Standard deviations on the selection composite for women also tended to be smaller than those for men. The distributions of the criterion, on the other hand, showed mean differences between men and women that ranged from 2.2 units for the Personnel Clerks to 4.8 units for the Aviation Supply Clerks, with women scoring higher in each case. Standard deviations for women were smaller than those for men for all but two courses.

Table 2
Means and Standard Deviations for Men and Women
on FCG and ASVAB Predictors

Training course/variable	Men		Women	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Administrative				
FCG	82.0	7.0	86.6	6.3
CL	106.9	14.0	117.1	10.2
AR	50.8	8.3	52.6	7.4
WK	53.2	7.1	56.9	5.6
AD	54.0	10.8	61.4	9.9
CA	12.3	3.3	13.6	2.9
Personnel				
FCG	89.2	5.4	91.4	4.3
CL	108.4	13.4	114.6	10.3
AR	52.0	7.7	52.3	7.2
WK	54.3	6.7	56.1	5.5
AD	53.6	11.3	59.3	11.1
CA	12.1	3.0	13.1	3.2
Unit diary				
FCG	82.0	6.8	86.8	7.1
CL	110.4	10.5	118.7	9.2
AR	53.2	7.3	54.4	7.6
WK	53.9	6.7	57.6	4.7
AD	53.9	10.6	61.2	10.6
CA	12.8	3.4	13.5	2.5
Basic supply stock				
FCG	82.1	7.6	85.4	8.1
CL	109.3	11.5	119.3	9.2
AR	53.7	6.6	54.9	6.9
WK	52.5	6.6	57.3	5.4
AD	57.4	10.9	61.0	10.7
CA	11.5	3.3	14.1	3.1
Personal financial records				
FCG	83.3	7.3	86.4	6.6
CL	112.8	10.2	119.7	8.2
AR	54.2	7.2	53.7	7.0
WK	56.3	5.3	58.3	4.4
AD	55.1	11.3	60.7	9.4
CA	12.1	3.1	14.4	2.8
Aviation operations				
FCG	86.2	5.8	90.8	4.6
CL	101.7	14.0	116.9	8.3
AR	49.1	7.4	53.6	6.1
WK	52.3	6.2	57.3	4.3
AD	52.8	11.7	59.8	9.7
CA	10.4	3.1	13.0	3.2
Aviation maintenance administration				
FCG	77.0	8.1	81.0	6.4
CL	103.8	13.2	117.4	8.6
AR	49.6	7.4	51.7	6.9
WK	52.8	6.1	57.2	5.2
AD	53.5	11.5	63.0	10.0
CA	11.0	3.3	13.0	3.0
Aviation supply				
FCG	81.6	8.1	86.3	7.0
CL	101.7	13.5	115.3	9.8
AR	49.2	7.6	51.4	7.2
WK	51.8	6.5	57.5	4.6
AD	53.1	11.8	60.7	10.9
CA	10.6	3.1	12.4	2.5
Communications center				
FCG	82.3	7.6	86.0	5.7
CL	106.0	11.7	114.4	9.4
AR	49.3	8.0	51.6	8.3
WK	52.4	7.1	56.3	5.8
AD	55.9	12.1	60.6	9.8
CA	11.8	3.4	12.6	3.1

Note. FCG = final course grade; ASVAB = Armed Services Vocational Aptitude Battery; CL = Clerical Composite; AR = Arithmetic Reasoning; WK = Work Knowledge; AD = Attention to Detail; and CA = Attentiveness Scale.

Complete findings regarding the regression analyses, significance tests, and Johnson-Neyman regions are presented in Dunbar and Novick (1985); however, certain specific results are worth noting. When the selection composite, CL, was the sole predictor, substantial differences between regression equations for men and women were observed—differences with respect to both slopes and intercepts. Using a significance level of .10, tests of the null hypothesis of parallel regression lines yielded significant slope differences for three training cohorts. In the cases of the two training programs with more than 100 women recruits, the common slope hypothesis was retained in one instance, $F(1,1283) = 2.29, p < .13$, for Administrative Clerks and rejected in the other, $F(1,951) = 3.18, p < .07$, for Supply Stock Clerks. For other training cohorts, a common observation was relatively sizeable slope differences, but observed significance levels well above .10. Follow-up tests of common intercepts yielded frequent rejections of the null hypothesis, the usual interpretation being that intercept differences were the rule rather than the exception in statistical comparisons of the within-group equations. We comment on such an interpretation below.

The results on the basis of the Johnson-Neyman regions of significance were generally similar to those on the basis of tests of slope and intercept differences. Regions of significance were determined for both training cohorts with large samples of female recruits, with scores on CL above 92 falling in the region for Administrative Clerks and scores on CL above 116, in the region for Supply Stock Clerks. Both of these regions included the average score for women recruits in the respective training program.

Interpretations of either the hypothesis tests for common slopes and intercepts or the Johnson-Neyman regions are complicated by the small number of female recruits in many of the training programs included in this study. Indeed, one reviewer commented that tests of common slopes typically require sample sizes larger than those used here for acceptable power to reject a false hypothesis. Drasgow and Kang (1984) documented this observation in a simulation study. From this point of view, lacking power to detect slope differences makes intercept differences a likely finding given the sequential nature of the tests proposed by Gulliksen and Wilks (1950). The Johnson-Neyman analyses are no less immune from this dilemma. Rather than confuse the issue and encourage overinterpretation of significance tests performed with these data, we instead adopt here a descriptive approach to the analysis of prediction differences.

Perhaps a clearer indication of the magnitude of male-female differences is provided in Table 3, which contains differences between criterion predictions for men and women at the mean of the predictor distributions for women and expressed in standard units for women. Negative differences here indicate the underprediction of women's performance on the criterion, whereas positive ones indicate overprediction. The values reported in Table 3 show the contrast between predictions for men and women to be quite uniform and dramatic, regardless of the training course or predictor variable considered. The male-female differences ranged from .14 standard deviations for CL in the Supply Stock course to 1.03 standard deviations

Table 3
Comparison of Predictions for Men and Women at the Mean for Women (Standard Units for Women)

Training course	Predictor					Multiple regression
	CL	AR	WK	AD	CA	
Administrative	-.40	-.64	-.55	-.64	-.68	-.51
Personnel	-.27	-.50	-.40	-.43	-.46	-.42
Unit diary	-.50	-.64	-.54	-.79	-.64	-.61
Supply stock	-.14	-.35	-.22	-.40	-.31	-.20
Personal financial records	-.18	-.50	-.42	-.36	-.35	-.45
Aviation operations	-.62	-.76	-.88	-.84	-1.03	-.70
Aviation maintenance	-.17	-.45	-.38	-.53	-.52	-.31
Aviation supply	-.17	-.55	-.37	-.53	-.63	-.23
Communications center	-.24	-.48	-.39	-.68	-.61	-.26
Weighted M^a	-.33	-.57	-.48	-.59	-.60	-.44

Note. CL = Clerical Composite; AR = Arithmetic Reasoning; WK = Word Knowledge; AD = Attention to Detail; CA = Attentiveness Scale.

^a Sample sizes for women used as weights.

for CA in the Aviations Operations course. A weighted average difference of about one-third of a standard deviation when the composite is used as sole predictor might call into serious question the hypothesis that women are not adversely affected by selection on the basis of a prediction equation for men or on the basis of a combined equation. Indeed, the average number of scaled score points that would have to be added to an average composite score for a woman to obtain the criterion score predicted by the corresponding equation is 9.88. An appropriate degree of caution is in order regarding the stability of male-female differences for training courses with a small number of women. However, the fact that these differences are in the same direction as differences found in the Gordon (1953) and Valentine (1977) studies cited earlier lends support to an argument that women may well be adversely affected by use of a combined equation for selection and suggests that some explanation for such differences be sought.

Alternative Predictive Composites

The use of ASVAB composites for predicting training success is obviously complicated by the kinds of within-group differences observed in this study. Users of ASVAB are faced with a dilemma encountered by many when issues of bias in selection arise. Separate algorithms for many training courses will no doubt lead to more accurate predictions within groups, but they make selection rules complex and sometimes difficult to justify from a policy standpoint. Gamache and Novick (1985) have discussed these problems in the context of educational selection and suggest several approaches for dealing with the problem. Clearly, a search for alternative composites, which ameliorate the problem of differential prediction without sacrificing mean squared errors of estimate, is suggested by the results considered to this point.

Several alternative regression models for Marine Corps clerical specialties were evaluated in an attempt to develop a more suitable selection index. Models in which selected ASVAB subtests were deleted occasionally yielded small decrements in the

amount of differential prediction observed for some training programs; these decrements typically involved slopes. As illustrated in one example below, however, alternative sets of ASVAB predictors typically increased, rather than decreased, prediction discrepancies between men and women.

The consistent differences between within-group regressions can have any number of explanations. Whatever the correct one, underprediction of women's performance suggests that the effect of some component of performance on the relevant criterion is specific to women. The exploratory analyses of alternative subsets of predictors made apparent that other ASVAB variables did not adequately measure such a component. Indeed, indicators of behavior that could effect success in a training program to a greater extent than they could high scores on selection tests (e.g., level of general education or previous experience in a particular area of training) are likely better at characterizing the component that is missing from the regression models considered thus far. Note that we were able to verify that race differences among the men and women included in these samples could not account for the kinds of prediction differences observed.

Although no indication of previous experience was available in the present validation context, Marine Corps recruits are routinely distinguished by whether they have received a high-school diploma. Prediction comparisons for men and women in a restricted sample of high-school graduates and the total sample (provided at the mean for women in Table 4) indicated that a substantial reduction in differential prediction could be achieved by using information about the receipt of a high-school diploma. The comparisons given in the column labeled AR/WK are based on within-group equations using the Arithmetic Reasoning and Word Knowledge subtests (components of the CL composite and one of the alternatives discussed) as joint predictors of FCG. Although the statistics in Table 4 clearly show that the AR/WK combination yielded differences for men and women of greater magnitude, the predictions for the sample of high-school graduates indicate smaller differences for both regression models shown. The most pronounced re-

Table 4
Comparisons of Predictions for Men and Women at the Mean for Women (Standard Units for Women)

Training course	Total sample		High-school graduates	
	CL	AR/WK	CL	AR/WK
Administrative	-.40	-.62	-.21	-.47
Personnel	-.27	-.34	-.19	-.28
Unit diary	-.50	-.81	-.25	-.50
Supply stock	-.14	-.35	-.14	-.28
Personal financial records	-.18	-.74	-.17	-.71
Aviation operations	-.62	-.74	-.59	-.70
Aviation maintenance	-.17	-.40	-.06	-.26
Aviation supply	-.17	-.53	.06	-.20
Communications center	-.24	-.33	-.05	-.16
Weighted <i>M</i> ^a	-.33	-.56	-.18	-.42

Note. CL = Clerical Composite; AR = Arithmetic Reasoning; WK = Word Knowledge.

^a Sample sizes for women used as weights.

ductions were observed in the Administrative, Unit Diary, and Communications Center cohorts when the composite was the sole predictor. Although women continued to be underpredicted, the average magnitude of that underprediction was reduced on consideration of high-school attainment. Again, because of the small sample sizes for women in several training courses, these results should be viewed as suggestive, rather than definitive, indications of the importance of nontest variables in reducing prediction differences between men and women.

Discussion

The results of this study present a challenge to military psychologists interested in the general problem of predicting success in training. Consistent differences between predictions were found for men and women in various Marine Corps clerical specialties, without any clear indication of the causes of those differences. The fact that a reduction in differential prediction was observed when differences between men and women in high-school attainment were removed is noteworthy. In other selection contexts, such as college admissions, one routinely observes that optimal prediction occurs with some weighted combination of admission test scores and a measure of high-school performance such as class rank or grade-point average. An argument suggested by the analyses in this study would advance the hypothesis that individual and group differences related to performance on the training criterion are explained in part by a performance composite much like high-school achievement. To ignore such information in the assignment of recruits to clerical specialties creates systematic group differences in expected performance. This is the kind of hypothesis that should be validated in other selection contexts and with sets of data containing a larger representation of women.

One referee suggested that the male-female variable may be confounded with socioeconomic status (SES) because of higher selectivity in the female population. A glance at the means and

standard deviations of CL (particularly from Table 2) confirms this selectivity, although it is unclear whether the restriction of range implied by sample statistics is not a characteristic of the female applicant pool itself rather than the result of differential selectivity into the validation sample. The finding that attainment of a high-school diploma reduced differential prediction is consistent with the hypothesis of SES confounding because of the known positive relation between SES and high-school attainment. This suggestion is not, however, damaging to the important conclusions of this study. Differences between regressions for men and women were apparent with the use of this version of ASVAB and the available personnel data, and, furthermore, these differences were reduced by the use of a prior education variable in prediction equations. Moreover, whatever the complex of variables is that caused the sex differences observed here, our results provide some disconfirmation of a strong statement that ASVAB composites and subtests make essentially the same predictions regardless of other characteristics of Marine Corps trainees. Were the present results inconsistent with previous findings from studies with large and balanced samples of men and women, this position would be more difficult to support on the basis of present findings alone.

A pragmatic view of the problem of differential prediction in employment or admissions testing would suggest a different strategy. Policy discussions in a variety of settings conclude that the use of separate equations for men and women, or for Blacks and Whites, is not a palatable solution to the problem of contrasting regressions for important subpopulations. Rather, palatable solutions are those that involve blocking on nonsensitive variables (such as education level) that correlate with sex, race, SES, and a future performance criterion. When this is feasible, the resulting selection procedure is less likely to create a relative disadvantage for members of any particular group, however membership in that group happens to be defined. Using information about the receipt of a high-school diploma in predicting training success in the military, as is done in current practice, is a useful first step in the employment of such variables.

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